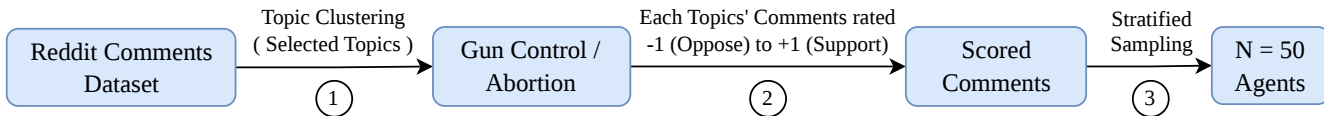
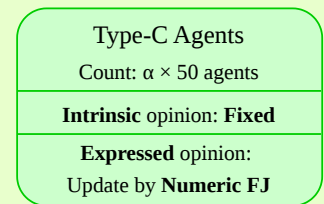


Stage 1 - Data Processing

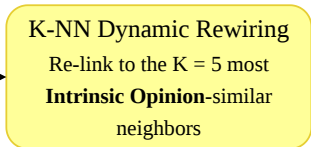


Stage 2 - H-COG Simulation

Splits 50 Agents into Two Types by α



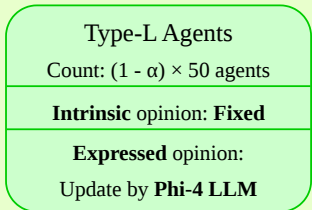
Opinion Updated
①



Network Updated
②

Repeat until convergence

③



Evaluated by

Polarization (Pz)

Variance of agents' **Expressed Opinions**, measuring how divided the opinions are.

Modularity (Q_norm)

How strongly the network clusters into like-minded communities, measured against a degree-matched random baseline.

Price of Anarchy (PoA)

Social Cost = Gap between own and neighbors' **Expressed Opinions** + Gap between own **Expressed** and **Intrinsic Opinion**
PoA = Actual Social Cost (Nash) ÷ Optimal Social Cost (Central Planner)